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Optimal Design of PEM Electrolyser with Integration of Renewable Power Supply for Forecasting of Green Hydrogen Production using Machine Learning

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Introduction

The demand for energy is ever increasing, and green hydrogen production emerges as pivotal for the advancement of the hydrogen economy in this context. Green hydrogen production is riddled with many scale up issues, with optimal design, prediction and process optimisation expected to be some of the most significant. The most common method for green hydrogen production is water splitting using electrolysis, where the electricity that powers this process comes from renewable resources. For the optimal design of an electrolyser, it requires the maximisation of the hydrogen production rate.

Aim:

This project aims to show the potential Machine Learning (ML) techniques have in addressing issues relating to green hydrogen production. Exploring the optimal design, prediction and process optimisation for a proton exchange membrane (PEM) electrolyser.

Objectives:

- To compare various ML models to determine which model best predicts hydrogen flow rate for PEM electrolysis
- To identify the optimal design of a PEM electrolyser for maximum hydrogen production
- To optimise process parameters of a PEM electrolyser for maximum hydrogen production
- To develop a model which integrates the optimised PEM electrolyser configuration with dynamic power inputs from a renewable energy source to predict the hydrogen production in a real-world scenario

Methods

Evaluation of best ML model for predicting hydrogen production:

- A dataset of 1062 datapoints was used for model training and testing, using an 80-20 split respectively
- The dataset contained different configurations of PEM electrolysers
- The ML models were trained to predict hydrogen production based on a set of input parameters
- 4 models were used for the comparison, which were as follows, K-Nearest Neighbours, Decision Trees, Random forest (RF), and Extreme gradient boosting.
- To evaluate the performance of the models, we used 3 performance metrics, root mean square error (RMSE) , mean absolute error (MAE) , and the correlation coefficient (R2).

Optimal design and process optimisation:

- For the optimal design, a feature importance analysis was used.
- For the optimal operating conditions of the electrolyser, bayesian optimisation was used
- The optimal design and operating conditions were fed back into the ML model creating an optimised electrolyser model

Hydrogen production simulation:

- Power production data collected from a 3.6 MW wind turbine was used an input into our optimised electrolyser model to predict hydrogen production based on renewable energy, using our design

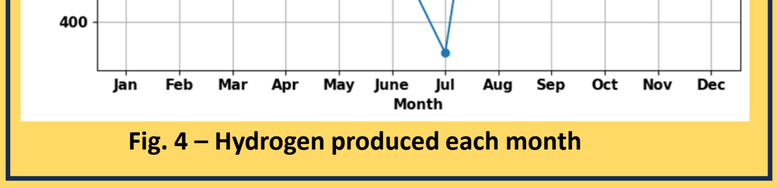


Fig. 1 - Proposed wind energy/hydrogen model

Results

Best ML model:

Random forest was found to be the best performing ML model for predicting hydrogen production for PEM technologies, with the lowest MAE, RMSE and highest R2 score.

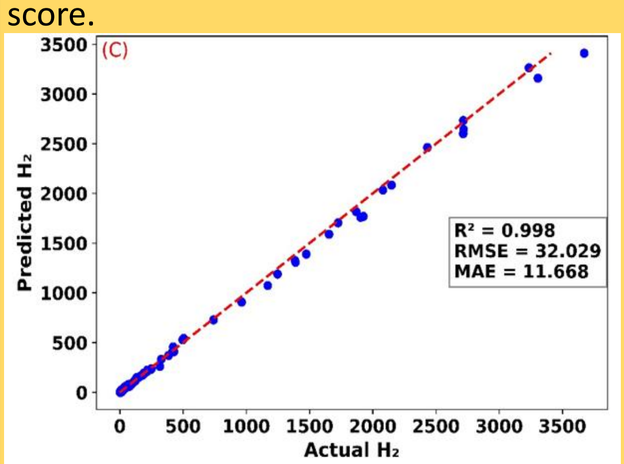


Fig. 2 – RF prediction accuracy plot

Optimisation results

- The optimisation algorithm returned the maximum bounded values available, except for pressure, electrode area, and cell temperature
- A further partial dependence analysis into the temperature on hydrogen flow rate revealed an optimal temperature range for PEM electrolysis

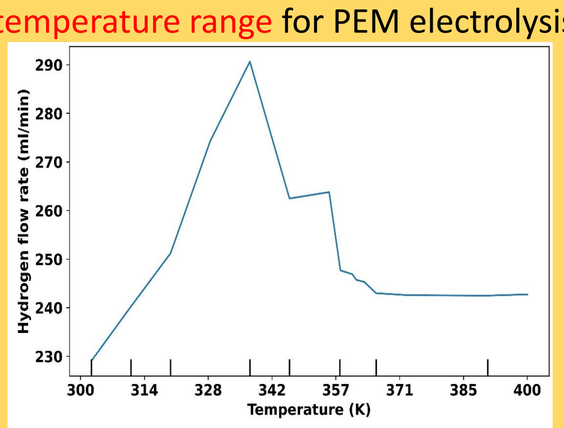


Fig. 3 – Partial dependence plot of hydrogen flow rate on temperature

Hydrogen production simulation results:

The simulation forecasted 10 tonnes of green hydrogen produced throughout the year, with production peaking in March and August.

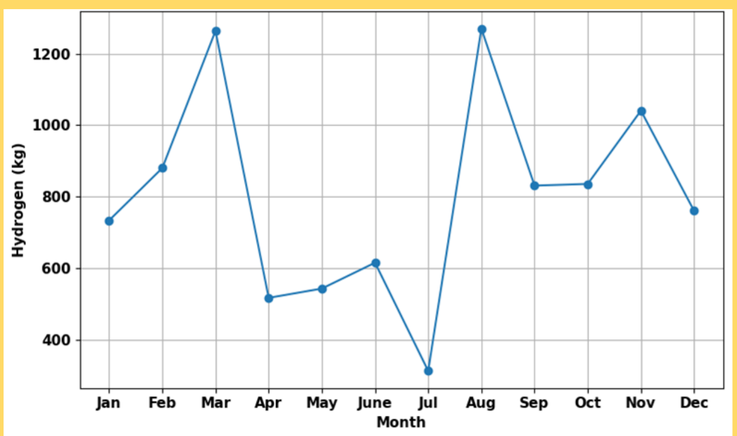


Fig. 4 – Hydrogen produced each month

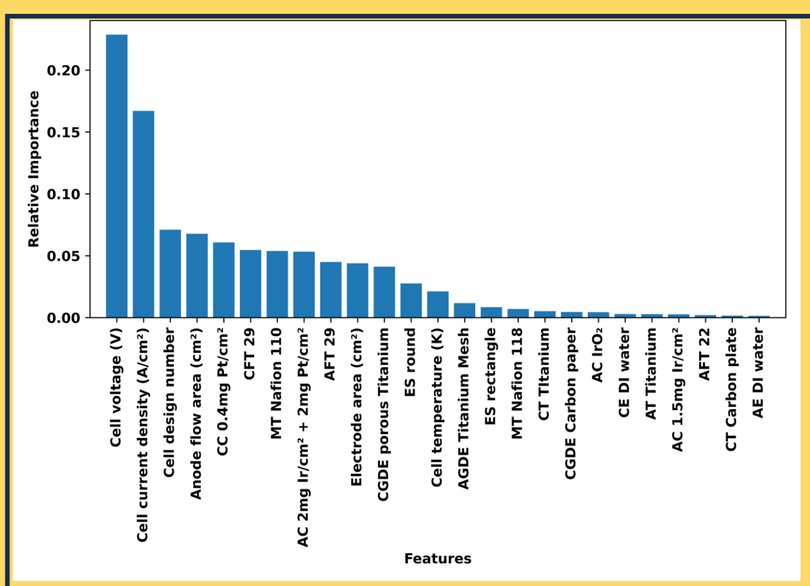


Fig. 5 – RF top 25 feature importance

Feature importance analysis:

- The feature importance analysis revealed that the Cell voltage was the most influential process parameter, followed by the cell current density
- This analysis also provided the optimal design for maximising hydrogen production

Conclusion

In conclusion, this study demonstrates the potential of ML techniques in enhancing green hydrogen production processes.

- Our feature importance analysis identified an optimal configuration for the electrolyser
- Our optimisation algorithm pinpointed the best operational conditions
- Partial dependence plots revealed complex relationships between hydrogen production rates and temperature variations

The successful application of these techniques for lab scale data is promising for future scaling of green hydrogen production